SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	Nabil The Goat
Project / Product Name	sabave.ai
Final Live Deployment URL	https://sabave.web.app
Codebase Repository URL	https://github.com/Joebilly112/sabaveai
Name (Leader)	Nabil Hilmi Tsaqifi
Email & phone (Leader)	nabiltsaqifi@gmail.com / +60 11 7244 6398

2. Project Description

[Maximum 250 words]

sabave.ai (save + brave) is a weather-triggered marketing automation module for Orbyte, built for Malaysian micro and small businesses. When haze arrives, walk-in sales fall: eateries in Kuala Lumpur and Shah Alam reported 20–35% drops in October 2026 (Malay Mail). Yet most owners do nothing, because one weather promotion costs about 90 minutes of their time to plan, design and post.

sabave.ai removes that busywork. It checks each outlet's PM2.5 forecast (Open-Meteo, Copernicus CAMS) and, when air quality is forecast to turn Unhealthy, opens an alert days ahead. Deterministic rules read the business's Brand Kit (products, prices, delivery, channels, budget cap and a never-say list) and propose campaign options, each with a plain-language reason and a suggested budget. Only then does generative AI (DeepSeek V4 Pro, server-side) write Bahasa Malaysia and English captions, checked against the never-say list, with a template fallback. Staff can draft and submit; only the Owner can approve or reject, enforced on the server. Approved campaigns are published through a Meta connector (simulated in the prototype), and every step is written to an activity log.

The same engine serves different businesses without code changes: a kopitiam gets a delivery push, while a laundromat that does not deliver gets pickup and indoor offers. At RM100 a month inside Orbyte, the sales recovered in one haze spell cover the subscription. The project targets SDG 8.2 and 8.3: more productive small businesses that protect their sales instead of cutting staff.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Overview. sabave.ai is a TypeScript pnpm monorepo with a React web app and an Express API, deployed on Firebase. Firebase Hosting serves the web app at https://sabave.web.app and rewrites every /api/** request to a Firebase Cloud Functions (2nd gen, Node 22) function named "api" that runs the Express server.

Frontend. React 19, Vite 7, TypeScript 5.9, Tailwind CSS 4 and shadcn/ui (Radix) components, wouter routing and TanStack Query. Screens: public site (Home, Solutions, Pricing), split login and sign-up, an 8-step onboarding wizard that builds the Brand Kit (including menu photos and team roles), a dashboard that leads with haze risk, a map and a 5-day forecast, campaign options, a draft editor, an approval screen, the activity log and a Brand Kit editor. A typed API client is generated from an OpenAPI specification with Orval, and payloads are validated with Zod.

Backend modules (Express 5): (1) Weather signal engine: pluggable signals; the haze signal fetches hourly PM2.5 for the outlet's coordinates from Open-Meteo (5 days, Asia/Kuala_Lumpur), takes the daily maximum and bands it Good <35, Moderate 35–55, Unhealthy 55–150, Very Unhealthy >150 µg/m³, with an offline forecast fallback. (2) Trigger engine: opens an alert when any day reaches Unhealthy and records the lead time in days. (3) Campaign rules: deterministic options from the Brand Kit (for example, a delivery push only if the business delivers) with budgets from severity, duration, an RM20 floor and the owner's monthly cap. (4) Copy generator (GenAI). (5) Approval service: role check, budget and never-say re-checks and a publish lock. (6) Social connector: a Meta interface that returns simulated post and ad IDs. (7) Activity log.

GenAI integration. The server calls DeepSeek's OpenAI-compatible chat-completions API (deepseek-v4-pro) and requests JSON. The prompt contains only business data; the model writes a headline, Bahasa Malaysia and English captions (≤60 words), hashtags, a call to action and a rationale, and is told not to invent prices, products or health claims or to choose the budget, option or publication. The server validates the JSON shape and the never-say list, retries once, and otherwise returns template copy and logs AI_FALLBACK. The same client summarises brand style from a poster or website (text only). The API key is a server-side environment secret and never reaches the browser.

Data structures. Business (id, type, city, lat/lon, team, Brand Kit: colours, tone, products {name, priceRM}, menu items, delivery, takeaway, channels, language, neverSay, monthlyAdBudgetCapRM,

approverUserId). WeatherAlert (signal, affected dates, peak, days ahead). CampaignDraft (option, copy, budget, channels, status draft → pending_approval → published | rejected, createdBy, approvedBy, rejectReason, publish result, history). ActivityLogEntry (id, ISO timestamp, actor, role, businessId, action, details). The prototype keeps these in an in-memory store; a Drizzle/PostgreSQL schema package is scaffolded and managed-database persistence is the next step.

Flow: Browser → Firebase Hosting (React app) → /api rewrite → Cloud Function (Express) → Weather signal engine → Open-Meteo → Trigger engine → Campaign rules (Brand Kit) → Copy generator → DeepSeek API (template fallback) → Approval service (Owner only) → Meta connector (simulated) → Activity log → dashboard.

Quality. Four Node test-runner suites cover alert triggering, budget rules, non-delivery options, never-say blocking, role enforcement (staff approval returns 403 and is logged), approval re-checks, AI fallback, input bounds and safe website fetching.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

During the physical sprint we turned a working prototype into a deployed, demo-ready product:

• Re-platformed: Replit became unavailable, so we moved development to Cursor and deployed the web app on Firebase Hosting with the Express API as a Cloud Function behind an /api rewrite, live at https://sabave.web.app.

• Real accounts and onboarding: split login and sign-up, Owner and Staff accounts, an 8-step onboarding wizard that builds the Brand Kit (menu photos, team roles) and a separate Brand Kit editor.

• Demo-ready test cases: per-account presets (kopitiam, laundromat, boutique) pre-fill onboarding, while new sign-ups start empty so judges can create their own business; the staff account joins the owner's shop and cannot approve; "Reset my demo" clears only that account. We removed the automatic seeding of every sample shop.

• Product and site: a dashboard that leads with haze risk and the next action, a dark map and compact forecast, a readable activity log, a public site (Home with a Malaysia globe, a Solutions page with cited haze evidence, single RM100 Pricing plan) and a new logo.

• AI: updated the DeepSeek client to deepseek-v4-pro with strict JSON output and validated template fallback.

Problem-solving from Phase 1 feedback: mentors asked for proof behind the impact claims, so time-to-campaign, steps and lead time are now measured from activity-log timestamps rather than estimated. We fixed a bug where the activity log read a cached HTML page as data, enforced and tested

that businesses without delivery never receive delivery campaigns, and replaced the Phase 1 plan (Next.js on Replit) with Vite, Express and Firebase when Replit was withdrawn.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	• Time per weather campaign: Phase 1 baseline ~90 minutes of owner time (team estimate; benchmark 8.4 h a week for 3.2 manual posts ≈ 2.6 h per post, Picmim 2026). In the final prototype, the system goes from alert to published in 20.5 seconds (activity-log timestamps: ALERT_TRIGGERED 2026-10-31T16:04:37.9Z → PUBLISHED 16:04:58.4Z, scripted run). That leaves only the owner's review. Even allowing a generous 10 minutes for the owner to read and approve, owner time falls by ≥89%, meeting the ≥85% target. • Lead time and steps: baseline 0 days (owners react after the haze). In the simulated haze scenario, the alert fired 1 day before the first Unhealthy day, against the ≥2-day target. That's partly met: the live Open-Meteo forecast covers 5 days, so real alerts can fire up to 4 days ahead, but this was not demonstrated in the test run. Manual steps: 7 → 4 (select option, edit/submit, approve, plus the alert). Banned wording published: unchecked → 0, blocked by rules and a server re-check.

Metric Category	Your Response
	• Not yet met: the ≥3 real-SME pilot from Phase 1; it is the first step of the scale-up plan.
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	• Database transaction log: every action (ALERT_TRIGGERED, DRAFT_GENERATED, SUBMITTED_FOR_APPROVAL, APPROVED, PUBLISHED) is written server-side with an ISO timestamp, actor and role. Time-to-campaign = t(PUBLISHED) − t(ALERT_TRIGGERED); steps = user actions between them; lead time = peak Unhealthy date − publish date, from the alert record. • Automated tests (Node test runner, 4 suites) verify alert thresholds, budget rules, no delivery option for non-delivery shops, staff approval rejected with 403 and logged as ACTION_DENIED, rejection requires a reason and publishes nothing, never-say blocking and AI-failure fallback. The same behaviour is shown live in the demo.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	• MSMEs produce 39.5% of Malaysia's GDP and 48.7% of employment (DOSM 2024), and the haze costs Malaysia an estimated RM20–50m a day (The Star, 14 Oct 2026). sabave.ai gives a kopitiam owner the early reaction a chain's marketing team has, protecting sales so owners do not cut staff (SDG 8.2, 8.3). • Illustrative owner value: RM1,300 daily sales × 22% dip × 3 days = RM858 at risk; recovering 15–20% (pilot hypothesis) returns RM129–172, more than the RM100 fee. • Scalable through partners: distributed inside Orbyte to SMEs already on the hub and through coffee-shop and hawker associations and POS/delivery platforms. The Brand Kit makes any walk-in business type work without code changes, and civic data such as official air-quality or flood alerts can plug in as new signals.

Metric Category	Your Response
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	• Modelled as an Orbyte plug-in module: it reads business profiles from the hub, exposes a typed REST API (OpenAPI), and offers a reusable social connector other modules can call. Live on Firebase (Hosting + Cloud Functions); running cost ≈ RM1.30 per outlet per month (AI ≈ RM0.03 per campaign; weather data free). • RM100 per month; after an assumed 30% Orbyte share, contribution ≈ RM68.70 (69%). Market: 136,453 F&B premises (DOSM 2023), starting with 38,474 in the Klang Valley; bottom-up projection of 770 outlets (≈ RM0.92m annual run-rate) and break-even in year 3.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Impact calculations (inputs from the activity log and Appendix A1 of the deck):

Time-to-campaign = t(PUBLISHED) − t(ALERT_TRIGGERED) = 16:04:58.4 − 16:04:37.9 = 20.5 s of system time (scripted run, 31 Oct 2026)

Owner time reduction, allowing 10 min for the owner's review: (90 − 10) ÷ 90 × 100 = 89% (target ≥85%)

Lead time = first Unhealthy day − alert date = 1 day (simulated forecast [45, 90, 165, 120, 50] μg/m³)

Owner value: 1,300 × 0.22 × 3 = RM858 at risk; 858 × 0.15–0.20 = RM129–172 recovered ≥ RM100 fee

Unit economics: 100 − 0.30 (AI, 10 × RM0.03) − 1.00 (hosting) − 30 (Orbyte share) = RM68.70 (69%)

Code (abridged from the repository):

```
// signals/hazeSignal.ts: PM2.5 bands; triggerLevel = 2 (Unhealthy)
function levelFor(value: number): SignalDay["level"] {
```

```
  if (value < 35) return 0;
  if (value <= 55) return 1;
  if (value <= 150) return 2;
  return 3;
}

// lib/campaignRules.ts: a delivery campaign only if the business delivers
if (business.brandKit.delivery) {
  options.push(option("delivery_push", "Bring your favourites home",
    "Promote delivery orders while customers spend more time indoors.",
    "You deliver, so a delivery push can keep orders coming while walk-ins fall."));
}

// lib/approvalService.ts: server-side approval gate
if (user.role !== "owner") {
  appendLog({ actor: user.name, role: user.role, businessId: draft.businessId,
    action: "ACTION_DENIED", details: { draftId: draft.id, decision: input.decision } });
  throw new CampaignError(403, "Only the Owner can approve or reject campaigns.");
}
if (draft.status !== "pending_approval" || publishing().has(draft.id))
  throw new CampaignError(409, "This draft is not awaiting approval.");
if (input.decision === "reject") {
    if (!input.reason?.trim()) throw new CampaignError(400, "A rejection reason is
required.");
  draft.status = "rejected"; draft.rejectReason = input.reason.trim();
  record(draft, user, "REJECTED", { reason: draft.rejectReason });
  return draft;                       // nothing is published
}
checkBudget(candidate, user); checkPublishable(candidate, user);   // re-check cap and
never-say
const result = await getConnector().publishPost(candidate, business);
record(draft, user, "APPROVED", { budgetRM: draft.budgetRM });
record(draft, user, "PUBLISHED", { ...result, budgetRM: draft.budgetRM });
```

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

With funding or an SDC partnership, sabave.ai moves from prototype to a paid Orbyte module in three stages.

Months 1–3, prove it: pilot with 3–5 F&B outlets in the Klang Valley, measuring revenue protected on haze days against days without a campaign, using POS or delivery-app sales. In parallel, move from the in-memory store to a managed database (Firestore, or PostgreSQL through the existing Drizzle schema), add scheduled forecast checks and complete Meta Graph API and WhatsApp Business approval so campaigns publish for real.

Months 3–12, launch inside Orbyte: sell at RM100 a month to SMEs already on the hub, avoiding cold customer acquisition, and add rain and heat signals so the product stays useful outside haze season. Partner with coffee-shop and hawker associations and POS and delivery platforms for distribution.

Years 2–3, scale: add Chinese and Tamil copy, more business types and flood or monsoon alerts. Our bottom-up model (each salesperson signs about 48 outlets a year, plus Orbyte and partner channels) reaches 770 outlets, about RM0.92m in annual run-rate revenue, and break-even in year 3, with a peak funding need of about RM75k. The addressable market is 136,453 F&B premises in Malaysia (DOSM 2023), and the same engine can expand to haze- and monsoon-affected markets such as Indonesia and Singapore.

The main risks are unproven willingness to pay and Meta API approval time; the pilot and a manual download-and-post fallback address both.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

N/A.

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React 19, React DOM	Code (library)	MIT
Vite 7, @vitejs/plugin-react	Code (build tool)	MIT
TypeScript 5.9	Code (language)	Apache-2.0
Tailwind CSS 4, tailwind-merge, tailwindcss-animate	Code (library)	MIT
Radix UI primitives (shadcn/ui components)	Code (library)	MIT
TanStack React Query; wouter	Code (library)	MIT; Unlicense
React Hook Form, @hookform/resolvers, Zod	Code (library)	MIT

Express 5, cors, cookie-parser	Code (framework)	MIT
pino, pino-http	Code (library)	MIT
Drizzle ORM, drizzle-zod (scaffolded)	Code (library)	Apache-2.0
Orval (API client generator)	Code (tool)	MIT
firebase-functions; Firebase Hosting and Cloud Functions	Code; cloud service	MIT; Google Cloud terms
Leaflet; react-leaflet	Code (library)	BSD-2-Clause; Hippocratic-2.1
react-globe.gl, three.js	Code (library)	MIT
Recharts, framer-motion, sonner, date-fns	Code (library)	MIT
lucide-react	Code (icons)	ISC
class-variance-authority	Code (library)	Apache-2.0
Replit Vite plugins and connectors SDK (development tooling)	Code (tool)	No licence stated (Replit)
Open-Meteo Air Quality API (Copernicus CAMS)	Dataset (API)	CC BY 4.0
DOSM Economic Census 2023 and MSME 2024 statistics	Dataset	DOSM open data terms
DeepSeek V4 Pro (deepseek-v4-pro, via API)	Model (pre-trained)	DeepSeek API Terms of Service

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

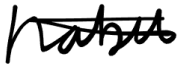
☐ Yes ☒ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

N/A.

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



10/31/2026